Quantifying Model, Label, and Methodological Contributions to Variance in Real-Time Sepsis Onset Prediction

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Master's Thesis · M.Sc. in Computing
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I. Introduction: Discrimination and Clinical Utility

- Sepsis is associated with about **11 million deaths a year**, nearly one fifth of deaths worldwide [1]. Each additional hour before antibiotics is associated with higher mortality [2].
- Wang et al. [3] reviewed **91** real-time sepsis prediction models:

Metric	Internal	External
Median AUROC	0.811	0.783
Median utility score	$+\mathbf{0.381}$	$-\mathbf{0.164}$

- Negative utility indicates performance below the no-alert strategy. AUROC measures ranking, not alert timing or clinical value, and therefore did not reflect the decline in utility.
- Deployment: Epic Sepsis Model v2, 227,091 encounters, AUROC 0.82–0.92, PPV 0.13–0.26 [4].

Model comparisons alone do not establish robustness or clinical utility.

II. Background: Limitations of the Conventional Approach

One label → one split → AUROC → model comparison

1. **Label definition:** applying Sepsis-3 differently produced 867–2,178 cases in the same database [5].
2. **Data split:** sepsis prediction showed the largest performance loss under temporal shift among the clinical tasks evaluated [6].
3. **Metric choice:** high AUROC can coexist with utility below the no-alert strategy [3].
4. **Clinical-action dependence:** labels may capture recognition and treatment practice, not physiological deterioration alone [7].

This thesis

It compares the **variation in performance** associated with model class, label definition, data split, prediction timing and evaluation metric, with **bootstrap confidence intervals** so that the magnitudes can be compared.

III. Research Question and Pre-Registered Hypotheses

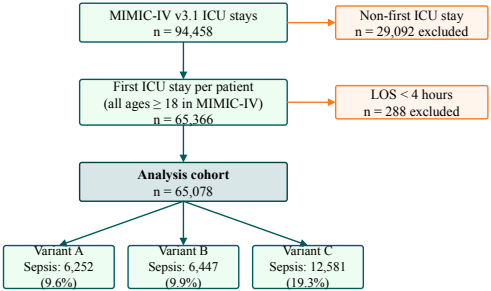
Research Question

How much of the variation in reported performance comes from the model, and how much from label, split, timing and metric?

H	Name	Pre-registered claim
H1	Label dominance	label spread $\geq$ model spread (AUROC)
H2	Coefficient instability	≥ 1 sign flip or ≥ 2-fold change
H3	Treatment leakage	pre-treatment AUROC lower by ≥ 0.03
H4	Split optimism	temporal AUROC ≥ 0.02 below random
H5	Metric divergence	utility falls more than AUROC externally
H6	Model-class null	GBT $-$ primary AUROC ≤ 0.02

Holm correction across H1–H6. Estimand: current-hour onset detection, not advance prediction.

IV. Materials: Data and Cohort



- **Development data:** MIMIC-IV v3.1 [8], single centre (BIDMC, Boston).
- **Cohort:** adults, first ICU stay, length of stay ≥ 4 h; **n** = $65{,}078$.
- **External validation:** eICU-CRD v2.0 [9]; **frozen models, no retraining**.
  over 200 US ICUs, $181{,}589$ unit stays
- **Person-hour data:** first 72 h; discharge and death modelled as competing events.
- **22 covariates;** 29 **model terms per outcome.**
  6 vitals, 5 labs, 2 vasopressor flags, 3 six-hour slopes, 6 test-order indicators

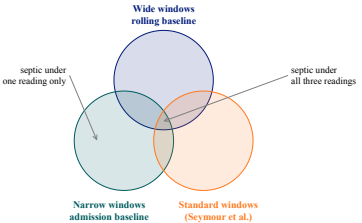
IV. Materials: Three Pre-Registered Sepsis-3 Label Definitions

ID	Name	ABX→cx	cx→ABX	SOFA−	SOFA+	Baseline	Septic stays
A	Narrow	24 h	24 h	24 h	12 h	Admission	6,252 (9.6%)
B	Seymour-Std	72 h	24 h	48 h	24 h	Admission	6,447 (9.9%)
C	Liberal	72 h	72 h	48 h	24 h	Rolling 24 h	12,581 (19.3%)

ABX: antibiotic; cx: culture; SOFA−/SOFA+: hours before/after suspected infection.

- All three: suspected infection + $\geq$ 2-point SOFA rise [10].
- **Variant B is primary.**
- Onset = t_{susp}; SOFA sets membership, not timing.

Similar prevalence, low agreement: $\kappa_{A,B} = 0.267$.



V. Methods: Models and Validation Architecture

Primary model: dynamic discrete-time competing-risks hazard

Multinomial pooled logistic regression over person-hours (sepsis / discharge / death), with a natural-spline baseline in elapsed hours [11], [12]. A small ridge penalty was used because the unpenalised model was completely separated.

Comparators: gradient-boosted trees (GBT, XGBoost), Cox regression, NEWS2, qSOFA and SIRS [13]. GBT used fixed hyperparameters and early stopping on a 15% stay-level training holdout.

Split	How	Why
Internal (primary)	Temporal split: train 2008–2016, test 2017–2019 [6]	Primary internal evaluation
Internal (secondary)	Random 75/25 patient-level split (seed 42)	Split optimism (H4)
Treatment-anchored	Hours before first culture or antibiotic [7]	Treatment leakage (H3)
External	eICU-CRD v2.0, no retraining [9]	Generalisability (H5)

All splits were patient-level, with no train–test overlap.

V. Methods: Metrics and Statistical Inference

Primary metric: adapted PhysioNet-style utility score [14]

$$\widetilde{U} = \frac{U - U_{\text{inact}}}{U_{\text{opt}} - U_{\text{inact}}} \qquad 1\text{: perfect forecaster,} \quad 0\text{: no-alert strategy,} \quad <0\text{: worse than no alerts}$$

- Rewards alerts from 6 h before onset to onset; penalises late and false alerts.
- Reported at the 0.5 threshold **and** at the best point on a 60-point alert-rate grid.
- Alert burden: false alerts per true alert; published benchmark: 1.4 [15].
- AUROC, AUPRC and calibration are reported alongside.

Inference and reporting

- Stay-cluster bootstrap, $B = 2{,}000$ replicates ($1{,}000$ for eICU-CRD).
- Holm correction across H1–H6 (FWER 0.05); Benjamini–Hochberg for exploratory analyses.
- Post-hoc amendments labelled explicitly, including the inference plan itself (Amendment 1).
- Reporting was guided by TRIPOD+AI [16], with risk-of-bias considerations informed by PROBAST+AI [17].

VI. Results: Internal Temporal Performance (Variant B)

Model	AUROC	95% CI	Utility (best)	Alert burden at best
Primary	0.7590	[0.7434, 0.7749]	0.025	47.3
GBT	0.7424	[0.7265, 0.7583]	0.031	51.8
Cox	0.6444	[0.6271, 0.6614]	0.000	∞
NEWS2	0.6415	[0.6248, 0.6589]	0.000	∞
SIRS	0.6114	[0.5950, 0.6281]	-0.035	135.8
qSOFA	0.6091	[0.5934, 0.6254]	-0.252	248.8

Temporal test set, 2017–2019: 548,827 person-hours and 1,040 onsets. Utility (best): maximised over the threshold grid; ∞: no alert was optimal. Benchmark burden: 1.4 [15].

- The paired comparison favoured the primary model over GBT by 0.0165 AUROC (H6). Both had higher AUROC than Cox, NEWS2, SIRS and qSOFA.
- Primary-model calibration slope: 1.055 (ideal = 1).
- At their best utility, the primary model and GBT required 47.3 and 51.8 false alerts per true alert, against a benchmark of 1.4.

VI. Results: AUROC Variation Across Analysis Choices

Factor varied (others fixed)	ΔAUROC	95% CI	H
Model class (label B, temporal)	0.1499	[0.134, 0.171]	n/a
Internal − external (primary, label B)	0.0630	[−0.0215, 0.1563]	H5
Label definition (primary, temporal)	0.0123	[0.003, 0.032]	H1
Random − temporal split (primary, label B)	−0.0165	[−0.0372, 0.0040]	H4
GBT − primary (label B, temporal)	−0.0165	[−0.0308, −0.0012]	H6

These are ranges or contrasts, not additive variance components. The model-class range covers six specifications and the label range three definitions, so the two are not directly symmetric. The GBT–Cox difference alone (0.7424 vs 0.6444) is about eight times the full label range.

- **Model class** produced the largest AUROC range.
- The external contrast was second largest but imprecisely estimated: its interval includes zero.
- **Label and split effects were small** in this analysis.

VI. Results: Confirmatory Hypotheses (Holm-Corrected)

H	Contrast	Est.	95% CI	p_{Holm}	Verdict
H1	Label spread − model spread	-0.1375	$[-0.1623, -0.1083]$	1.000	NOT confirmed
H2	Unstable covariates (count)	6	n/a	n/a	Descriptive
H3	Pre-treatment onsets (count)	0	n/a	n/a	Definitional
H4	Random − temporal AUROC	-0.0165	$[-0.0372, 0.0040]$	1.000	NOT confirmed
H5†	Internal − external AUROC	0.0630	$[-0.0215, 0.1563]$	0.551	Not evaluable as pre-registered; substituted AUROC contrast inconclusive
H6	GBT − primary AUROC	-0.0165	$[-0.0308, -0.0012]$	$1.000^{\ddagger}$	H6 not rejected

† **Post-hoc substitute estimand.** ‡ **H6 places $|\theta| \leq 0.02$ in the null, so this p-value tests H6 itself; non-inferiority is read from the CI.**

H1: model spread 0.1499 exceeds label spread 0.0123. **H6**: upper limit $-0.0012 < +0.02$.

VI. Results: Why H3 Was Not Evaluable

Status: (NOT EVALUABLE AS PRE-REGISTERED)

Structural reason

Onset is the *later* event of the antibiotic–culture pair, so it cannot precede treatment: the window is empty **by definition**.

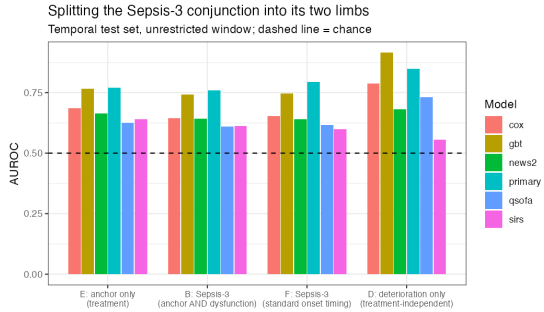
Var	Label	Pre-treatment person-hours	Onsets	AUROC (primary / GBT)
B	Seymour-Std	179,003	0	not estimable
F	Sepsis-3 Early-Onset	178,303	71	0.8081 / 0.8018
D	Deterioration*	147,299	1,362	0.8498 / 0.9360

- **F** (post-hoc): standard onset $\min(t_{\text{susp}}, t_{\text{SOFA}})$ moves 1,738 of 6,447 onsets earlier, 71 into the test window.

*D's SOFA components overlap the model inputs, so its AUROC is not an independent result.

VI. Results: Post-Hoc Decomposition of the Sepsis-3 Label

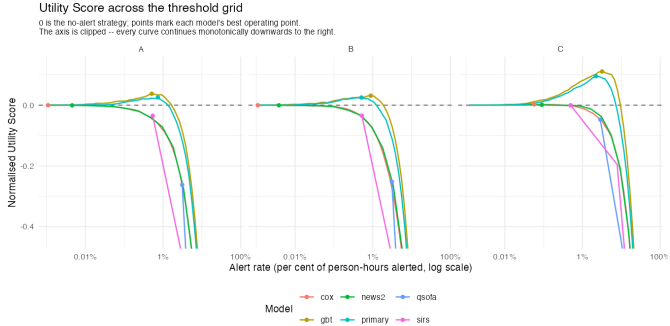
Labels D, E and F are post-hoc protocol amendments: (EXPLORATORY, POST-HOC)



- **E (Anchor-Only)** removes SOFA but keeps B's onset hour in 100.0% of B's cases: SOFA affects membership, not onset time.
- Anchor-attributable fraction $\phi_{\textbf{anchor}}$: primary 1.041 $[0.990, 1.097]$; GBT 1.099 $[1.037, 1.164]$.
- **D (Deterioration)** removes treatment timestamps; agreement with B is low ($\kappa = 0.057$). Five of its six SOFA components are model inputs, so its AUROC is not an independent result.
- **F (standard onset timing):** primary AUROC $0.7590 \rightarrow 0.7951$; the GBT change was small and not statistically detectable ($\Delta = 0.0033$, $q = 0.559$).

For the primary model, onset timing had a larger effect than removing SOFA. No corresponding timing effect was detected for GBT.

VI. Results: Clinical Utility Across Alert Thresholds



- At the 0.5 threshold, the primary model issued no alerts: utility 0.000.
- Highest Variant B utility: 0.031 **(GBT)**, at 51.8 **false alerts per true alert** (benchmark: 1.4 [15]).
- Highest under any pre-registered variant: 0.111 (GBT, C).
- Thresholds were selected on the test set, so these maxima are optimistic.

VI. Results: External Validation Limited by Label Observability

	MIMIC-IV	eICU-CRD			
Var	Primary model	Primary model	GBT	NEWS2	Events
A	0.7713	0.6913	0.7106	0.5866	27
B	0.7590	0.6960	0.6815	0.5259	29
C	0.7695	0.7370	0.6877	0.5074	73

eICU microbiology-covered subset; frozen models, no retraining.

- Culture data covered 1.55**% of eICU cohort stays**, leaving 29 Variant B events, below the 100-event interpretability floor.
- H5 substitute, internal − external AUROC $= 0.0630$ $[-0.0215, 0.1563]$: the point estimate is lower externally, but the interval includes zero, so the contrast is inconclusive. Its sources were not decomposed.
- Antibiotic-only ($6{,}140$ events) and Variant D ($52{,}535$ events) analyses had more events but measured different targets; both showed lower external discrimination.

VI. Results: Coefficient Instability and Subgroup Analysis

H2: coefficient instability (descriptive)	
Covariates meeting the rule	6 of 22 (2 sign, 5 magnitude; overlap)
Still unstable after estimation uncertainty	3**, all test-order indicators**
Ablation: drop 8 action covariates (post-hoc)	1 of 14 unstable; ΔAUROC -0.0076
Subgroup analysis (exploratory)	
Subgroup AUROC contrasts with ≥ 100 events	6; **none significant** (BH)
Label sensitivity above the White reference	*Unknown*, *Unable to obtain*

After accounting for uncertainty, instability remained only in test-order indicators [18]; label-sensitivity differences were limited to missingness categories [19].

VII. Discussion: Main Findings and Limitations

Main findings

1. **Onset timing:** H3 not evaluable; standard timing moves $1{,}738$ onsets earlier.
2. **Definition:** prevalence 9.6–19.3%, A–B $\kappa = 0.267$, yet AUROC range only 0.0123.
3. **Utility:** 51.8 false alerts per true alert at best (benchmark 1.4).

Improving discrimination alone cannot resolve limitations introduced by outcome membership, onset timing and alert burden.

Limitations

1. Current-hour onset detection, not advance warning.
2. A–C share one treatment anchor, so part of their spread is definitional.
3. H6 is non-inferiority against one untuned GBT.

Also: single centre; eICU 1.55% culture coverage (29 events); post-hoc variants D and F.

VIII. Conclusion: Contributions and Future Work

C1	Model class gave the largest AUROC range; the label-definition range was only 0.0123 (A–B $\kappa = 0.267$).
C2	The pre-registered onset rule empties the pre-treatment window; standard timing places 71 test-set onsets there.
C3	Best Variant B utility 0.031 at 51.8 false alerts per true alert.
C4	3 covariates remained unstable after accounting for uncertainty, all test-order indicators.

Future work: treatment-independent labels; lagged predictors; prospective multi-centre validation.

The main effect of label choice was on cohort membership and onset timing rather than on AUROC. Model choice affected AUROC more, and utility stayed close to the no-alert strategy.

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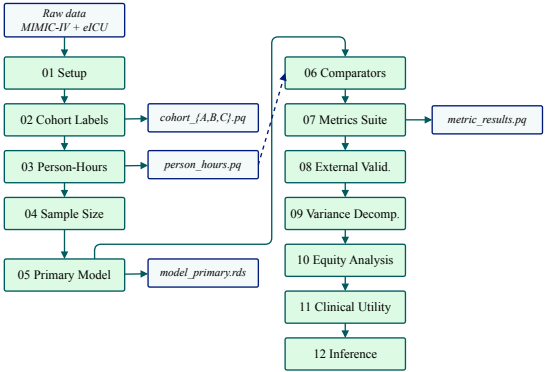
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Appendix: Reproducible Analysis Pipeline



- One orchestration script runs all 12 stages.
- One configuration file defines every path, identifier, window, threshold and seed.
- Every reported value is generated directly from the pipeline's result tables; none is transcribed by hand.
- Frozen run: `frozen-2026-08-31`.
- Code:

`github.com/ahadarif007/sepsis-variance-decomposition`

Thank you

Questions and discussion

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Code: `github.com/ahadarif007/sepsis-variance-decomposition`